Prayas JEE AIR (2025)

Advanced

Weekly Test 22

DURATION : 180 MINUTES | **DATE : 21/12/2024** | **M. MARKS : 198**

ANSWER KEY

PHYSICS		CHEMISTRY		MATHEMATICS	
1.	(A)	19.	(D)	37.	(B)
2.	(B)	20.	(C)	38.	(C)
3.	(D)	21.	(B)	39.	(B)
4.	(B)	22.	(D)	40.	(A)
5.	(C)	23.	(B)	41.	(B)
6.	(A)	24.	(C)	42.	(A)
7.	(A, B, C, D)	25.	(A, B, C)	43.	(A, C, D)
8.	(A, B, C)	26.	(A, B, D)	44.	(B, D)
9.	(A, B, C)	27.	(A, B)	45.	(B, C, D)
10.	(B, D)	28.	(A, C, D)	46.	(C,D)
11.	(C, D)	29.	(B, C)	47.	(A, B, C)
12.	(A, D)	30.	(C, D)	48.	(A, D)
13.	(8.00)	31.	(22.40)	49.	(2.00)
14.	(6.00)	32.	(1.73)	50.	(9.00)
15.	(2.00)	33.	(375.00)	51.	(196.00)
16.	(4.00)	34.	(4.50)	52.	(32.00)
17.	(2.00)	35.	(6.00)	53.	(30.00)
18.	(216.00)	36.	(57.39)	54.	(0.00)

PART-I (PHYSICS)

1. **(A)**

$$S_A = v_0 t - \frac{1}{2}\mu_A g t^2$$

$$S_B = v_0 t - \frac{1}{2}\mu_B g t^2$$

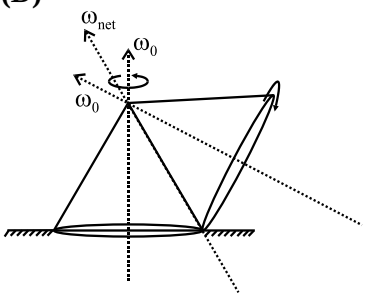
$$S_A - S_B = L - l \quad \Rightarrow t = \sqrt{\frac{2(L-l)}{(\mu_B - \mu_A)g}}$$

2. **(B)**

$$P_0 - \frac{2T}{r} + \rho g h = P_0 - \frac{2T}{2r}$$

$$\Rightarrow T = \rho g h r$$

3. **(D)**



$\omega_{net} = 2\omega_0 \cos(\alpha)$

$$= 2 \times 2 \times \frac{\sqrt{3}}{2}$$

$$\omega_{net} = 2\sqrt{3}$$

$$(KE) = \frac{1}{2} I_\alpha (\omega_{net})^2$$

I_α = Moment of inertia of cone about its slant axis having semi apex-angle α.

$$I_\alpha = \frac{MR^2}{4}\left(1 + 3\cos^2(\alpha)\right) = \frac{13MR^2}{16}$$

$$(KE)_{net} = \frac{1}{2}\left(\frac{13 \times 2 \times (1)^2}{16}\right) \times 4 \times 3 = \frac{39}{4}\text{J}$$

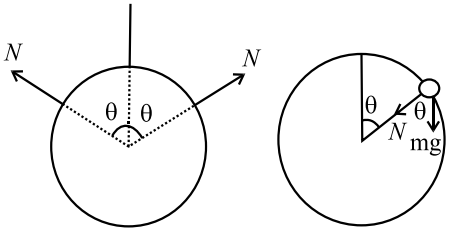
4. **(B)**

$$\frac{V_1}{V_2} = \frac{L_1 i_1 + M i_2}{L_2 i_2 + M i_1}$$

$$= \frac{L_1 i_1 + \sqrt{L_1 L_2} i_2}{L_2 i_2 + \sqrt{L_1 L_2} i_1} = \sqrt{\frac{L_1}{L_2}}$$

5. **(C)**

Balance the force in vertical direction



T becomes zero when

$2N\cos\theta = Mg$ ….(1)

$$mgR(1 - \cos\theta) = \frac{1}{2}mv^2 \quad ....(2)$$

$$mg\cos\theta + N = \frac{mv^2}{R} \quad .....(3)$$

From equation (1), (2) & (3) we get

$6m\cos^2\theta - 4m\cos\theta + M = 0$

$$\cos\theta = \frac{1}{3} \pm \frac{1}{3}\sqrt{1 - \frac{3M}{2m}}$$

6. **(A)**

Tension in the string

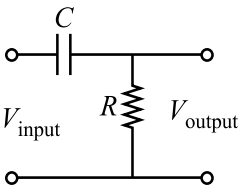
$T = mg - F_B = 3\rho V g - \rho V g = 2\rho V g$

Balancing torque

$kxR/2 = TR$

$\Rightarrow x = 2T/k = 4\rho V g/k$

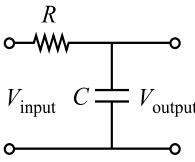
7. **(A, B, C, D)**



(A) $V_{out} = V_{in}\frac{R}{z}$

$$V_{out} = \frac{v_0 R}{z}\sin\omega t$$

$$v_M = \frac{v_0 R}{z}; z = \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$



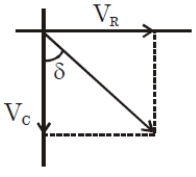
(B)

$$v_{out} = v_{in}\frac{X_c}{z}$$

$$v_{out} = \frac{X_c v_0}{z}\sin\omega t, \qquad v_M = \frac{X_c v_0}{z}$$

(C) $P_{avg} = \frac{v_0 I_0}{2}\cos\phi$,

(D)



$$\tan\delta = \frac{R}{x_C}$$

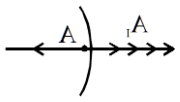
$\tan\delta = \omega RC$

8. **(A, B, C)**

$A \Rightarrow OD = -ut$

$ID = ?$

$$\frac{1}{-ut}+\frac{1}{ID}=\frac{1}{-f}$$

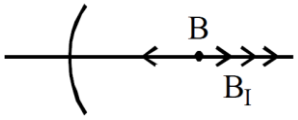


$$\frac{1}{ID}=\frac{1}{ut}-\frac{1}{f}$$

$$ID=\frac{utf}{f-ut}=\frac{f}{\frac{f}{ut}-1}$$

$B \Rightarrow OD = -(2f - ut)$

$ID = ?$



$$\frac{1}{-(2f-ut)}+\frac{1}{ID}=\frac{1}{-f}$$

$$\frac{1}{ID}=\frac{1}{2f-ut}-\frac{1}{f}$$

$$ID=\frac{(2f-ut)f}{-f+ut}=\frac{(2f-ut)f}{-f+ut}$$

$$|ID|=\frac{(2f-ut)f}{f-ut}=f\left(1+\frac{f}{f-ut}\right)=f\left(1+\frac{1}{1-\frac{ut}{f}}\right)$$

9. **(A, B, C)**

Also $W=\frac{P\propto V^{2/3}}{\frac{nR\Delta T}{1-x}}\frac{P_0V_0}{RT_0}=\frac{(4P_0)(8V_0)}{RT}$

$\Rightarrow \quad T = 32T_0$

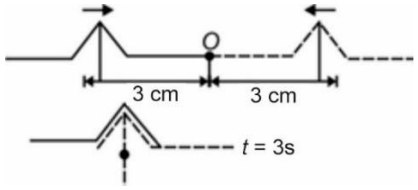
$$=\frac{nR(31T_0)}{1+\frac{2}{3}}=\frac{93RT_0}{5}$$

$$\Delta Q(1)\left[\left(\frac{3}{2}R\right)+\left(\frac{3}{5}\right)R\right](31T_0)$$

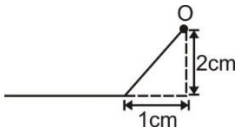
$$=\frac{21}{10}R(31T_0)=65.1RT_0$$

10. **(B, D)**

The formation of the reflected pulse from a free support is similar to the overlap of two pulses of same nature travelling in opposite directions in same phase as shown below.



Therefore, net displacement of all points will become twice as shown below:



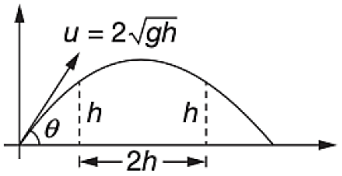
11. **(C, D)**

$a_{BA} = 3a_0$ {using constraint relations}

$$a_{BG}=\sqrt{a_0^2+9a_0^2+6a_0^2\cos\theta}$$

$$a_0\sqrt{10+6\cos\theta}$$

12. **(A, D)**



$$h=2\sqrt{hg}(\sin\theta)t-\frac{1}{2}gt^2$$

$$\Rightarrow \quad t^2-4\sqrt{\frac{h}{g}}(\sin\theta)t+\frac{2h}{g}=0$$

$$\Rightarrow \quad t=\frac{4\sqrt{\frac{h}{g}}\pm\sqrt{16\left(\frac{h}{g}\right)\sin^2\theta-\frac{8h}{g}}}{2}$$

$$\Rightarrow \quad t=\frac{4\sqrt{\frac{h}{g}}\pm\sqrt{\frac{8h}{g}}\sqrt{2\sin^2\theta-1}}{2}$$

$$\Rightarrow \quad t_1=\frac{4\sqrt{\frac{h}{g}}-\sqrt{\frac{8h}{g}}\sqrt{2\sin^2\theta-1}}{2}$$

$$\Rightarrow \quad t_2=\frac{4\sqrt{\frac{h}{g}}+\sqrt{\frac{8h}{g}}\sqrt{2\sin^2\theta-1}}{2}$$

$$\Rightarrow \quad (t_2-t_1)=\sqrt{\frac{8h}{g}}\sqrt{2\sin^2\theta-1}$$

Further horizontal distance travelled during this time is $2h$

$$\Rightarrow \quad 2h=2\sqrt{hg}\cos\theta(t_2-t_1)$$

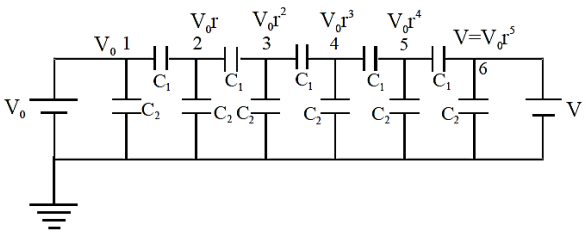
$$\Rightarrow \quad 2h=2\sqrt{hg}\cos\theta\sqrt{\frac{8h}{g}}\sqrt{2\sin^2\theta-1}$$

Solving we get $\cos\theta=\frac{1}{2}$

$\theta = 60°$ with horizontal or $30°$ with vertical

$$(t_2-t_1)=2\sqrt{\frac{h}{g}}$$

13. **(8.00)**



$C_1 (V_0 r - V_0) + (V_0 r) C_2 + (V_0 r - V_0 r^2) C_1 = 0$

$$r - 1 + \frac{rC_2}{C_1} + r - r^2 = 0$$

$$\frac{4r}{3} + 2r = r^2 + 1$$

$$3r^2 + 3 - 10r = 0$$

$$r = 3 \text{ or } \frac{1}{3} \Rightarrow V = V_0 r^5$$

$$V_0 3^5 \text{ or } \frac{V_0}{3^5}$$

14. (6.00)

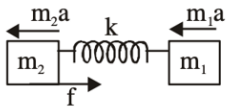
$$-\frac{Qdq}{A\varepsilon_0} = -\frac{1}{2} m v_0^2 \qquad \text{...(i)}$$

$$-\frac{Qdq}{A\varepsilon_0} = \frac{1}{2} m \left(v^2 - (2v_0)^2 \right) \qquad \text{...(ii)}$$

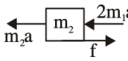
$$\Rightarrow v = \sqrt{\frac{6qQd}{\varepsilon_0 ms}}$$

15. (2.00)

Viewed from frame of plank



[initial condition]



[at the time of maximum compression]

$\because 2m_1 a + m_2 a = 8\text{N}$ and $f_{max} = \mu\text{N} = 10$ N

$\Rightarrow$ Friction is static

$$kA = m_1 a \Rightarrow A = \frac{m_1 a}{k} = 2 \text{ cm}$$

16. (4.00)

Flux : $\phi = B.\pi r^2$

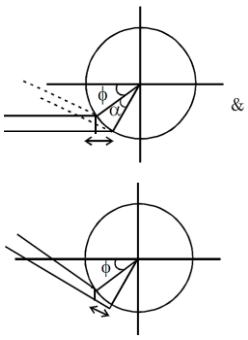
Induced emf : $\varepsilon = \left| \frac{d\phi}{dt} \right| = 2\pi Br \frac{dr}{dt} = 2\pi Brv.$

Induced current : $I = \frac{\varepsilon}{R} = \frac{2\pi Brv}{R}$

Rate of work done = Rate of heat dissipation

$$= I^2 R = \frac{4\pi^2 B^2 r^2 v^2}{R}$$

17. (2.00)



$$\left| r\alpha \sin\phi - r\alpha \cos(\pi - \theta - \frac{\pi}{2} - \phi) \right| = \lambda$$

$$\sin\phi - \sin(\theta + \phi) = \frac{\lambda}{r\alpha}$$

$$r\alpha = \frac{\lambda}{|\sin\phi - \sin(\theta + \phi)|}$$

18. (216.00)

Let m be the mass of the container initial temperature of container is

$T_i = (227 + 273) = 500$ K

And final temperature of container is

$T_f = (27 + 273) = 300$ K

Now (Heat Gained by the Ice Cube) = (Heat lost by the container)

$$m_{\text{ice}} L_{\text{ice}} + m_{\text{ice}} c_{\text{water}} \Delta T = m \int_{T_i}^{T_f} c dT$$

$\Rightarrow (0.1)(8 \times 10^4) + (0.1)(10^3)(27)$

$$= m \int_{500}^{300} (A + BT) dT$$

$$\Rightarrow 10700 = m \left(AT + \frac{BT^2}{2} \right)\Bigg|_{500}^{300}$$

After substituting the values of A and B and the proper limits, we get

$$= m = \frac{107}{216} \text{ kg}$$

PART-II (CHEMISTRY)

19. (D)



20. (C)

$[Cr(NH_3)_3Cl_3]$ can show Mer-fac isomerism

$[Cr(en)_3]Cl_3$ can show optical isomerism

21. (B)

	$H_2(g)$	+	$I_2(g) \rightleftharpoons 2HI(g)$
Initial mol :	1	1	0
mol at eqlm:	$(1-x)$	$(1-x)$	$2x$
eqlm conc.	$\frac{(1-x)}{V}$	$\frac{(1-x)}{V}$	$\frac{2x}{V}$

Now, equivalents of hypo solution used = equivalents of I_2 present at equilibrium

$0.1(10 \times 1) = (1-x) \times 2$

$x = 0.5$

$$\text{So, } K_c = \frac{(1)^2}{\left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right)} = 4$$

So, K_c for the reaction

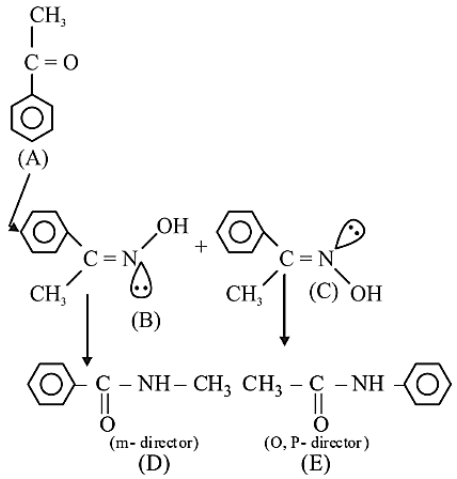
$$HI(g) \rightarrow \frac{1}{2}H_2(g) + \frac{1}{2}I_2(g)$$

will be $\left(\frac{1}{4}\right)^{\frac{1}{2}} = \frac{1}{2} = 0.5$

22. (D)

All lanthanoids are non-radioactive except Pm.

23. (B)



24. (C)

NO_2 will give HNO_2 and HNO_3.

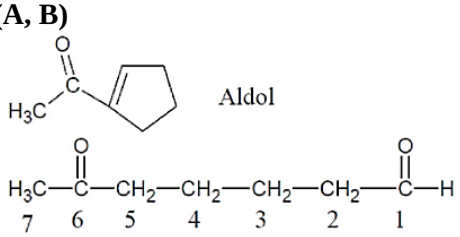
25. (A, B, C)

Lone pair of N-atom participating in resonance will be less basic

26. (A, B, D)

$[CoF_6]^{3-}$ is a high spin complex.

27. (A, B)



28. (A, C, D)

Formula of ideal solid 'x' $= AB_3$

$$\Delta T_f = i \times k_f \times m$$

$$0.2976 = 1.6 \times 1.86 \times \frac{0.60 \times 1000}{m_x \times 100}$$

$$m_x = 60\,g/mol$$

$$r_{A^{3+}} + r_{B^-} = \frac{a}{\sqrt{2}} \therefore a = 2\ A$$

$$\text{Density (d)} = \frac{60 \times 1}{6 \times 10^{23} \times 8 \times 10^{-24}} = 12.5\,g/cm^3$$

Total number of unit cells = Total number of molecules of $AB_3 = \frac{0.60}{60} \times 6 \times 10^{23} = 6 \times 10^{21}$

If there are 12% cationic vacancies then it means that formula of defected solid would be

$$A_{0.88}B_{3.00} \text{ So, } \%A^{4+} \text{ ions } = \frac{36}{88} \times 100 = 40.9\%$$

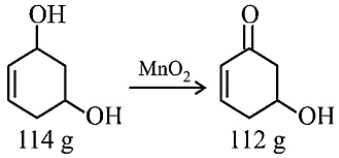
29. (B, C)

$Cr^{3+}Al^{3+}$, Sn^{2+}, Pb^{2+} form soluble complexes with NaOH

30. (C, D)

Electron withdrawing group increases reactivity towards ArSN

31. (22.40)



$$\text{Weight of product (P)} = \frac{112}{114} \times 22.8 = 22.40\,g$$

32. (1.73)

Specific conductance of AgCl = Specific conductance of solution - specific conductance of H_2O

$= (3.32-1.5) \times 10^{-6} = 1.82 \times 10^{-6}\,\Omega^{-1}cm^{-1}$

For saturated solution of sparingly soluble salt

$$\lambda_{eq} = \lambda_{eq}^{\infty}$$

$$\therefore \quad \lambda_{eq}^{\infty} = \frac{1000 \times 1.82 \times 10^{-6}}{S} = 138.3$$

$$\therefore \quad S = 13.159 \times 10^{-6}\ M$$

$$= 13.16 \times 10^{-6}\ M$$

$$K_{sp} = S^2 = 173.1856 \times 10^{-12}$$

$$= 1.73 \times 10^{-10}$$

33. (375.00)

Reaction is first order w.r.t B, so half-life is constant if B is not changed

Reaction is second order w.r.t A so half-life becomes half on doubling the conc. of A

312.5 + 62.5 = 375

(y + x)

34. (4.50)

y = 0, x = 9.

35. (6.00)

$$\overset{+7}{MnO_4^-} + \overset{+2}{S_2O_3^{2-}} \rightarrow \overset{+4}{MnO_2} + \overset{+6}{SO_4^{2-}}$$

Equivalents of MnO_4^- =equivalents of SO_4^{2-}

Moles of MnO_4^- × n-factor = moles of SO_4^{2-} × n-factor

$8 \times 3 = X \times 4 \ \therefore \ X = 6$

8 moles of MnO_4^- produces 6 moles of SO_4^{2-}

36. (57.39)

$$\Delta S^\circ = \sum S^\circ_{product} - \sum S_{reactant}$$

$\Delta S° = 2(238.9) – [151.3 + 221.0]$

$= 477.8 – 372.3$

$=105.5 \text{ J mol}^{-1} \text{ K}^{-1}$

$\Delta G° = \Delta H° – T\Delta S°$

$\Delta G° = 25.7 \times 10^3 – (298 \times 105.5)$

$= –5739$

$\therefore \ \frac{5739}{100} = 57.39$

PART-III (MATHEMATICS)

37. (B)

$$\left|\bar{c} - \bar{b} + 2\bar{a}\right|^2 = |\bar{c}|^2 + |\bar{b}|^2 + 4|\bar{a}|^2 - 2\bar{b}\cdot\bar{c} - 4\bar{a}\cdot\bar{b} + 4\bar{c}\cdot\bar{a}$$

$\because \ \bar{b}\cdot\bar{c} = 0, \ \frac{\bar{b}\cdot\bar{a}}{|\bar{a}|} = \frac{\bar{c}\cdot\bar{a}}{|\bar{a}|}$

$$\left|\bar{c} - \bar{b} + 2\bar{a}\right|^2 = 1 + 1 + 4 = 6$$

38. (C)

$$\frac{2x^3y}{\left(1-y^2\right)^2}\frac{dy}{dx} - 1 + \frac{y^2x^2}{\left(1-y^2\right)} = 0$$

$$\Rightarrow \quad \frac{2y}{\left(1-y^2\right)^2}\frac{dy}{dx} + \left(\frac{y^2}{1-y^2}\right)\frac{1}{x} = \frac{1}{x^3} \quad .....(1)$$

Put $\frac{y^2}{1-y^2} = t$

$$\Rightarrow \quad \frac{2y}{\left(1-y^2\right)^2}\frac{dy}{dx} = \frac{dt}{dx} \qquad .....(2)$$

$\therefore \quad \frac{dt}{dx} + \frac{t}{x} = \frac{1}{x^3}$, This is a linear differential equation.

$\Rightarrow$ I.F. = x

$$\therefore \quad t\cdot x = \int x\cdot\frac{1}{x^3}dx + c$$

$$= -\frac{1}{x} + c$$

$$\Rightarrow \quad \frac{x^2y^2}{1-y^2} = (cx - 1)$$

39. (B)

$$f'(x)f(x) \geq \left(f(x)\right)^4 + 1$$

$$\Rightarrow \frac{2f'(x)f(x)}{\left\{\left(f(x)\right)^2\right\}^2 + 1} \geq 2$$

$$\Rightarrow \int_0^a \frac{2f'(x)f(x)}{\left\{\left(f(x)\right)^2\right\}^2 + 1} \geq 2\int_0^a 1dx$$

$$\Rightarrow \left|\tan^{-1}\left(f(x)\right)^2\right|_0^a \geq 2a \Rightarrow \frac{\pi}{3} - \frac{\pi}{4} \geq 2a$$

40. (A)

$$= 9\left[(10-1) + \left(10^2-1\right) + \left(10^3-1\right) + ... + \left(10^n-1\right)\right]$$

$$= \left[10^{n+1} - 9n - 10\right]$$

$a = 1, b = –9, c = –10$

41. (B)

$n(x_1) = 3, x_1 = 2, 3, 4$

$4 \leq x_2 < x_3 \leq 8 \Rightarrow$ from 4, 5, 6, 7, 8

Selecting any 2

$\Rightarrow n(x_2, x_3) = {}^5C_2 = 10$ and $n(x_4) = 2, \ x_4, = 0, 5$

$\Rightarrow$ Number of such 4 digit numbers $3 \times 10 \times 2 = 60$

42. (A)

Total formed numbers that begin with a odd digit

$= {}^5C_1 \cdot {}^8P_4 = 5(8)(7)(6)(5)$

Total formed numbers that end with a odd digit

$= {}^5C_1 \cdot {}^8P_4 = 5(8)(7)(6)(5)$

Total formed number that begin with an odd digit and also end with an odd digit
$= {}^5C_2 \cdot 2! \cdot {}^7P_3 = 5 \cdot (4)(7)(6)(5)$
Thus total formed numbers that begin with an odd digit or end with an odd digits is equal to $5 \cdot 7 \cdot 6 \cdot 60$
Total formed numbers $= {}^9P_5 = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5$
Thus, required probability $= \frac{5}{6}$

43. (A, C, D)

We have $\log_{\cos x}\left(\frac{\sqrt{3}}{2}\sin x\right) - \log_{\cos x}(\tan x) = 2$

$\Rightarrow \log_{\cos x}\left(\frac{\sqrt{3}}{2}\cos x\right) = 2$

$\Rightarrow \frac{\sqrt{3}}{2}\cos x = \cos^2 x. \Rightarrow \cos x\left(\cos x - \frac{\sqrt{3}}{2}\right) = 0$

$\therefore x = \frac{\pi}{6}\ \left(x = \frac{\pi}{2} \text{ is rejected}\right)$

$x = 30°$

44. (B, D)

$I_1 = \int_0^{\infty} \frac{x\,dx}{1+x^4}$; Put $x^2 = t$; $2x\,dx = dt$

$\Rightarrow \quad I_1 = \frac{1}{2}\int_0^{\infty} \frac{dt}{1+t^2} = \frac{1}{2}\tan^{-1} t\Big]_0^{\infty} = \frac{\pi}{4}$

$I_2 = \int_0^{\pi/4} \frac{x\,dx}{\cos x(\cos x + \sin x)} \qquad .....(1)$

$I_2 = \int_0^{\pi/4} \frac{\left(\frac{\pi}{4} - x\right)}{\left(\frac{1}{\sqrt{2}}\right)(\cos x + \sin x)\left(\frac{2}{\sqrt{2}}\cos x\right)} dx$

.....(2)

$\Rightarrow$ Adding (1) and (2), we get

$2I_2 = \frac{\pi}{4}\int_0^{\pi/4} \frac{dx}{\cos x(\cos x + \sin x)}$

$= \frac{\pi}{4}\int_0^{\pi/4} \frac{\sec^2 x\,dx}{(1+\tan x)} = \frac{\pi}{4}\ln(1+\tan x)\Big]_0^{\pi/4}$

Hence $I_2 = \frac{\pi}{8}\ln 2$

$I_3 = \int_1^{\frac{1+\sqrt{5}}{2}} \frac{1+\frac{1}{x^2}}{x^2 - 1 + \frac{1}{x^2}} \ln\left(1 + x - \frac{1}{x}\right) dx$

$= \int_1^{\frac{1+\sqrt{5}}{2}} \frac{1+\frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 1} \ln\left(1 + x - \frac{1}{x}\right) dx$

Put $x - \frac{1}{x} = t \Rightarrow \left(1 + \frac{1}{x^2}\right)dx = dt$

If $x = 1$, $t = 0$; $x = \frac{\sqrt{5}+1}{2}$, $t = 1$ (verify)

$\Rightarrow I_3 = \int_0^1 \frac{\ln(1+t)dt}{1+t^2}$; Put $t = \tan\theta$

$\Rightarrow I_3 = \int_0^{\pi/4} \ln(1+\tan\theta)d\theta = \frac{\pi}{8}\ln 2$

$I_4 = \int_0^1 \frac{\sin^{-1} x}{x} dx$

Put $\sin^{-1} x = t \Rightarrow x = \sin t \Rightarrow dx = \cos t\,dt$

$\Rightarrow I_4 = \int_0^{\pi/2} \underset{\text{(I)}}{t} \cdot \underset{\text{(II)}}{\cot t}\ dt$ (I.B.P.)

$= \left(t \cdot \ln(\sin t)\right)_0^{\pi/2} - \int_0^{\pi/2} \ln(\sin t)dt$

$= 0 - \lim_{t\to 0^+} t\ln(\sin t) - \left(-\frac{\pi}{2}\ln 2\right)$

$\left(\text{As } \int_0^{\pi/2} \ln(\sin t)dt = -\frac{\pi}{2}\ln 2\right)$

$= -\lim_{t\to 0^+} \frac{\ln(\sin t)}{\frac{1}{t}}\left(\frac{\infty}{\infty}\right) + \frac{\pi}{2}\ln 2$

$= -\lim_{t\to 0^+} \frac{\cot t}{\frac{-1}{t^2}} + \frac{\pi}{2}\ln 2$

$= \lim_{t\to 0^+}\left(\frac{t}{\tan t} \times t\right) + \frac{\pi}{2}\ln 2 = \frac{\pi}{2}\ln 2$

45. (B, C, D)

Let $\alpha, \beta, \gamma, \delta$ are roots of equation.

$\therefore \quad \alpha + \beta + \gamma + \delta = -l \qquad ....(1)$

$\sum\alpha\beta\gamma = -n \qquad ...(2)$

$\alpha\beta\gamma\delta = 25 \qquad ...(3)$

$\therefore \quad \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = -\frac{n}{25}$

$\because \ (\alpha+\beta+\gamma+\delta)\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}\right) \geq 16$

$\therefore \quad ln \geq 400$

46. (C, D)

Let P be (x_1, y_1)

So equation of tangent at P will be

$xx_1 + yy_1 - (y + y_1) - 1 = 0$

Comparing with $x + y = 3$

$(x_1, y_1) \equiv (1, 2)$

So R & Q will be

$\left(1 \pm \frac{2\sqrt{2}}{3}\cos\frac{3\pi}{4}, 2 \pm \frac{2\sqrt{2}}{3}\sin\frac{3\pi}{4}\right)$

$\Rightarrow \quad Q\left(\frac{5}{3},\frac{4}{3}\right) \& R\left(\frac{1}{3},\frac{8}{3}\right)$

Let Q lies on $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

So tangent at P is

Comparing with $x + y = 3$

$a^2 = 5, b^2 = 4 \Rightarrow e_1 = \frac{1}{\sqrt{5}}$

and R lies on $\frac{x^2}{a_1^2}+\frac{y^2}{b_1^2}=1$

So tangent at $\frac{x}{3a_1^2}+\frac{8y}{3b_1^2}=1$

Comparing with $x + y = 3$

$\Rightarrow \quad a_1^2=1, b_1^2=8$

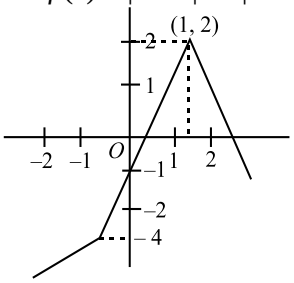
$\Rightarrow \quad e_2=\sqrt{\frac{7}{8}}$

$\Rightarrow \quad e_1^2+e_2^2=\frac{43}{40} \& e_1e_2=\frac{\sqrt{7}}{2\sqrt{10}}$

$\left|e_1^{-2}-e_2^{-2}\right|=\frac{27}{7},\ e_1^{-2}.e_2^{-2}=\frac{40}{7}$

47. (A, B, C)

Let $f(x) = |x + 1| - 2|x - 1|$



48. (A, D)

$x_1 + y_1 = -8$

$x_2 + y_2 = -8$

$x_3 + y_3 = -8$

$A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$

$$A(\Delta ABC)=\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}=0$$

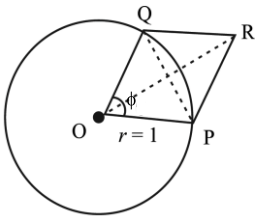
$\therefore$ Points A, B, C are collinear.

49. (2.00)

Area of the parallelogram outside the circle

= Area of parallelogram – Area of sector OPQ

$=\left(\frac{1}{2}\sin\phi\right)2-\frac{1}{2}\phi=\sin\phi-\frac{\phi}{2}=f(\phi)$



50. (9.00)

$x = {}^nC_6 (3^{1/3})^{n-6} (4^{-1/3})^6$

$y = {}^nC_{n-6} (3^{1/3})^6 (4^{-1/3})^{n-6}$

and $\frac{x}{y}=\frac{1}{12} \Rightarrow (12)^{(n-12)/3}=(12)^{-1} \Rightarrow n=9$

51. (196.00)

$AB = B$

$\Rightarrow (a - 1)p + bq = 0$ and $cp + (d - 1)q = 0$

$(a - 1)(d - 1) - bc = 0$

$\Rightarrow ad - bc = \det(A) = 49$

$\det(2A) = 2^2 \det(A) = 4 \times 49 = 196$

52. (32.00)

$zw = |z|^2 \Rightarrow |z|\,|w| = |z|^2$

$\Rightarrow \quad |w| = |z|$

Let $z = re^{i\theta} \Rightarrow w = re^{-i\theta}$

$\Rightarrow \quad w=\bar{z} \Rightarrow |z-\bar{z}|+|z+\bar{z}|=8$

$|x|+|y|=4 \Rightarrow \text{Area}=4\left(\frac{1}{2}\cdot 4\cdot 4\right)=32$

53. (30.00)

$$I_1 + I_2 = \int_0^{\pi} f(x).\sin x\,dx + \int_0^{\pi} f''(x)\sin x\,dx$$

$$= f(x).-\cos x\Big|_0^{\pi} + \int_0^{\pi}\cos x.f'(x)dx$$

$$+ \sin x\, f'(x)\Big|_0^{\pi} - \int_0^{\pi}\cos x.f'(x)dx$$

$\Rightarrow \quad f(\pi) + f(0) = 5(given)$

$\Rightarrow \quad f(0) = 5 - f(\pi) = 5 - 2 = 3$

54. (0.00)

$f(x) = \sin(\tan^{-1}(\cos(\cot^{-1}x)))$

$x \in R \Rightarrow f(x) \in \left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if